



Examiners' Report

Summer 2026

Pearson Edexcel GCE
in Mathematics (9MA0)
Paper 32 Mechanics

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General Comments

The paper proved to be reasonably accessible with the majority able to make good attempts at all six questions. Generally, time was not a factor but there were some who did not attempt the last part of the last question.

Question 1 and the first three parts of question 2 provided the opportunity for learners to settle into the paper and gain some confidence. There were some excellent scripts but there were also some where the standard of presentation could be improved. This, in some cases, made it difficult for examiners to follow the working. Learners should try to spread their work out as this will make it easier to read.

In calculations, the numerical value of g which should be used is 9.8. Final answers should then be given to 2 (or 3) significant figures – more accurate answers may not gain the relevant accuracy mark, including fractions but exact multiples of g are usually accepted.

There were a number of printed answers to show on this paper (in 2(a), 3(a), 5(b) and 6(a)), and learners *must* ensure that they show sufficient detail in their working to warrant being awarded all of the marks available and that they end up with **exactly** what is printed on the question paper, with no errors in the working. It is evident that many learners understand these requirements, clearly establishing necessary equations, including at least one line of working and re-arranging their final answer, when necessary, to match the printed answer. However, there were a number of cases where it was similar but not exactly as printed. It is important that learners complete all steps to show the result exactly as printed in order to be awarded full marks.

In all cases, as stated on the front of the question paper, learners should show sufficient working to make their methods clear to the examiner and correct answers without working may not score all, or indeed, any of the marks available.

If a learner runs out of space in which to give their answer than they are advised to use a supplementary sheet, and it is **important** that the learner indicates whereabouts in the script the extra working is going to be done.

Comments on individual questions

Question 1

Part (a)

The majority identified the correct method and completed the calculation accurately with most using $v^2 = u^2 + 2as$ but a few found the time first and then used it to find the height.

A few learners thought that the projection was at an angle as opposed to vertically upwards and some learners set $s = 0$, indicating a misunderstanding of the quantity being sought. Common errors seen were in signs, use of $g = 9.81$ and replacing $2g$ by 4.9 instead of 19.6 .

Part (b)

This part proved to be more challenging than the first part.

The most successful solutions used the equation $v = u + at$ directly to determine the required time. However, many learners did not appreciate that the initial and final velocities were in opposite directions and so could not gain the accuracy mark due to having a sign error in their equation.

Some chose to split the motion into two stages, first finding the time to reach the highest point then the time from the highest point to the ground and adding the two together. Although this approach was less efficient, those who used it generally applied it correctly and arrived at the correct answer. The downside of a longer method is that it increases the likelihood of arithmetical errors.

A number of learners calculated only the time taken for the object to travel from its highest point to the ground, rather than the required total time.

A significant number of learners gave their answer as a fraction and so the final accuracy mark was not awarded.

Question 2

Overall, this question was answered well. Most learners made a strong start and scored well in parts (a), (b), and (c) with very few using *suvat* equations in the second and third parts. Part (d) proved to be a strong discriminator.

Part (a)

The majority successfully substituted $t = 8$ into the given velocity vector and proceeded to set up a quadratic equation in c , scoring two of the three marks. However, many did not consider or reject the negative solution and so the final accuracy mark was not awarded.

Part (b)

This part was answered very well, with most learners successfully differentiating both components of the vector function to earn the method mark and in most cases the accuracy mark too. There were slips when subtracting 1 from the powers, some gave their answer as a column vector and a few integrated.

Part (c)

Learners generally attempted the integration with confidence and many successfully found a correct expression for the displacement vector. The main source of error was in dealing with the constant of integration.

Part (d)

Common unsuccessful responses resulted from learners using their displacement/position vector from part (c) instead of the given velocity vector to determine the direction of motion.

For those who correctly identified the need to use the velocity vector, a common difficulty was how to convert their calculated angle into a three-figure bearing and some also did not consider the direct instruction to round their final bearing to the "nearest degree."

Question 3

Part (a)

Those learners who followed the instruction in the question to resolve vertically usually produced the required equation needed to find the normal reaction. The majority of responses showed their trigonometric substitutions clearly and were awarded all the marks for this part of the question. Some understood the need to resolve vertically but incorrectly labelled R as being vertically upwards on their diagram rather than perpendicular to the plane and as a result were not awarded any marks other than for the use of $F = \mu R$. Others took the normal reaction to be $0.4g \cos \theta$, without accounting for the contribution of the horizontal force. The vast majority were able to access the B mark for the use of $F = \mu R$, even if it was seen in their work for part (b). A large number of learners resolved parallel and perpendicular to the plane but labelled this as a 'vertical' resolution. These were only able to access the B mark in this part of the question, as they had not followed the instruction given to resolve in a vertical direction.

This was a 'Show that..' question and a minority of learners did not explicitly show the substitution of the relevant trigonometric values and were not awarded the final accuracy mark.

Part (b)

Many learners were able to make a successful attempt at this part of the question, regardless of their resolutions in part (a). The majority were able to score some marks for their attempt at finding an equation in X and θ only, with many writing down correct equations even if they continued from their work in part (a). Some resolved parallel and perpendicular to the plane, then eliminated R to reach an equation in X and θ only while others simply resolved horizontally and substituted in the given value for R . The most common reason for marks not being awarded in this part was from learners giving an answer of 21.56, not recognising that they should round to 2 or 3 significant figures after using $g = 9.8$. Other errors occasionally seen were the omission of g , sine and cosine confusion or errors in algebraic manipulation of their equation(s), but these tended to be rare.

Question 4

This question is a good discriminator, with part (b) being more challenging than part (a).

Part (a)

This part was generally done well with the vast majority writing down an equation of motion for the whole system. Common errors included having either extra g 's or missing g 's, including the reaction force in the equation and errors in algebraic manipulation.

Part (b)

This part was generally answered less well. Almost all learners who were successful, wrote down an equation of motion for the block only. There were many learners who were not clear which object they were focusing on for whom there was an inconsistency between the forces appearing in their equation and the ' ma ' term. Many thought that the tension in the cable also acted on the block

and again there were missing or extra g 's. Some obtained a correct value of the reaction using a correct method but then went on to add it to or subtract it from another force, and gave a different, incorrect value as the final answer, and were not awarded all the marks for an incorrect method.

Question 5

Parts of this projectile question were well answered by most learners. Most recognised the need to use *suvat* equations for both the horizontal and vertical motion of the projectile, with most learners correctly resolving the initial velocity to find appropriate components. Following the use of $g = 9.8 \text{ m s}^{-2}$ it is important that final answers are given to 2 or 3 significant figures as directed on the front of the exam paper. Not doing this is penalised once per complete question and similarly for the use of $g = 9.81 \text{ m s}^{-2}$.

Part (a)

Most responses successfully used $0 = (28 \sin \alpha) t - \frac{1}{2} g t^2$ to find the total time of flight. Occasional errors were made in substituting in values, calculating $0.5g$ or omitting a t from one of the terms. Learners should be advised to always quote a *suvat* formula before applying it since the method mark will generally be available even if there is a subsequent slip in the substitution of values.

Some learners found the time value needed to reach the maximum height using $0 = (28 \sin \alpha) - gt$, but many then did not double their t value to obtain the time for the whole motion. Virtually all used the correct vertical component of the initial velocity in whichever *suvat* method they chose.

The most common error seen involved the accuracy of the final answer. A large proportion of learners did not give their answer to 2 or 3 significant figures as required following the use of $g = 9.8 \text{ m s}^{-2}$, with the fraction $\frac{24}{7}$ being a common

answer. A small number gave the required time as 3.42 rather than 3.43 after doubling the rounded $t = 1.71$ (time to reach greatest height).

Part (b)

Most learners were able to score the first three marks for this part of the question by using the horizontal component of velocity and the given displacement to obtain the time taken to reach the point A. Some proceeded to use this t value to calculate a vertical displacement rather than a vertical velocity component. This often led to no further progress with learners attempting to find $\tan \theta$ with displacements rather than velocities. However, a small number did successfully use this calculated displacement to find the required vertical velocity rather than the more direct approach of $v = (28 \sin \alpha) - gt$. The time from the highest point rather than from O was also used on occasion. Responses that found the correct horizontal and vertical components of velocity at A generally proceeded to establish the given answer for $\tan \theta$.

Part (c)

Most learners who had successfully completed part (b) were able to find the required value of V , the speed of the stone as it passed through A. Some who had not calculated the vertical component of the velocity previously did so here, generally correctly. Most learners used Pythagoras to find the speed, although solutions using the given value of $\tan \theta$ and just one of the components, often the horizontal, were seen. Occasionally only one component of velocity was stated rather than the required speed. Again, the answer was required to 2 or 3 significant figures following the use of $g = 9.8 \text{ m s}^{-2}$. Those who gave it to more figures (often 22.79) or left it as a surd achieved the mark here if they had already been penalised in part (a) for over-specification.

Part (d)

Many learners stated a correct limitation of the model with the most common responses involving either the spin or the dimensions/shape of the stone. The most common incorrect answers involved either the mass/weight of the stone or a statement that the ground may not be level/horizontal. The latter was not stated as a modelling assumption in the question and so was not relevant here. One limitation was required; those who chose to offer more than one risked not gaining the mark if one of those answers was incorrect including adding extra wrong information, for example 'it ignores the dimensions of the stone such as its weight'.

Question 6

Many learners found this question challenging, with a significant number making no progress. Among those who did attempt the question, some spent time calculating angles within and outside the trapezium $ABDC$, although these calculations were generally unnecessary and did not contribute towards an efficient solution. Much successful solutions were obtained by learners who took the given $\tan \beta = \frac{15}{8}$ and worked out exact values for $\sin \beta$ and $\cos \beta$.

Part (a)

Most answers scored at least M1A1, but many were not awarded the second accuracy mark due to not showing explicitly the value of $\sin \beta$. Since the question is a 'Show that', examiners were expecting both $\sin \beta$ ($15/17$) and T ($17\text{ mg}/6$) to be explicitly substituted into the moments equation and to see the 17's being cancelled. Only a small number of responses were affected by an incorrect number of terms because they did not identify the correct perpendicular distance for the moment of the tension or for not including 'a' in their terms. It was quite common to come across blank responses or just the values of $\sin \beta$ and $\cos \beta$ written down.

Part (b)

This part proved to be more demanding. Many learners resolved vertically and/or took moments about B and they managed to do these well. However, a much smaller number realised that they needed to also resolve horizontally in order to establish the pair of simultaneous equations required for a complete solution. There were some sin/cos errors, but most managed at least one equation and it was common to see a score of M0A0M1A1M0A0 in part (b).

A minority of learners resolved vertically, took moments about B and/or took moments about G , thereby creating multiple equations which all led to $T \sin \alpha = \frac{1}{2} mg$.

Some learners found the two equations which were needed but a common difficulty was knowing how to eliminate α once the equations had been formed. For those that did, there were some who used the numerical value of α but most found the exact value of $\sin \alpha$ and substituted. It was more common to see this method rather than the use of $\cos^2 \alpha + \sin^2 \alpha = 1$, although if this was attempted it was generally successful.

Those learners who chose to find the value of the angle α were not awarded the final mark as they were unable to find the exact expression for the tension as printed.

Very rarely, learners chose to use the cosine rule on a triangle of forces.

A notable proportion of learners did not attempt part (b), reflecting the level of challenge presented by this question.